Module 5: CORROSION AND ITS PREVENTION

Corrosion of metals is defined as the spontaneous destruction of metals in the course of their chemical, electrochemical or biochemical interactions with the environment. Thus, it is exactly the reverse of extraction of metals from ores.

Example: Rusting of iron
A layer of reddish scale and powder of oxide ($Fe_2O_3.xH_2O$) is formed on the surface of iron metal.
A green film of basic carbonate [$CuCO_3 + Cu(OH)_2$] is formed on the surface of copper, when it is exposed to moist-air containing carbon dioxide.

CONSEQUENCES (EFFECTS) OF CORROSION:
The economic and social consequences of corrosion include
i) Due to formation of corrosion product over the machinery, the efficiency of the machine gets failure leads to plant shut down.
ii) The products contamination or loss of products due to corrosion.
iii) The corroded equipment must be replaced
iv) Preventive maintenance like metallic coating or organic coating is required.
v) Corrosion releases the toxic products.
vi) Health (eg., from pollution due to a corrosion product or due to the escaping chemical from a corroded equipment).

CAUSES OF CORROSION:
The metals are extracted from their metallic compounds (ores). During the extraction, ores are reduced to their metallic states by applying energy in the form of various processes. In the pure metallic state, the metals are unstable as they are considered in excited state (higher energy state). Therefore as soon as the metals are extracted from their ores, the reverse process begins and form metallic compounds, which are thermodynamically stable (lower energy state). Hence, when metals are used in various forms, they are exposed to environment, the exposed metal surface begin to decay (conversion to more stable compound). This is the basic reason for metallic corrosion.

$$\text{Metal} \underset{\text{Metallurgy-Reduction}}{\overset{\text{Corrosion-Oxidation}}{\rightleftharpoons}} \text{Metallic Compound} + \text{Energy}$$

CLASSIFICATION OR THEORIES OF CORROSION

Based on the environment, corrosion is classified into

A. **Dry or Chemical Corrosion** **B. Wet or Electrochemical Corrosion**

A. DRY or CHEMICAL CORROSION:

This type of corrosion is due to the direct chemical attack of metal surfaces by the atmospheric gases such as oxygen, halogen, hydrogen sulphide, sulphur dioxide, nitrogen or anhydrous inorganic liquid, etc. The chemical corrosion is defined as the direct chemical attack of metals by the atmospheric gases present in the environment.

Example: (i) Silver materials undergo chemical corrosion by Atmospheric H_2S gas .

(i) Iron metal undergo chemical corrosion by HCl gas.

TYPES OF DRY or CHEMICAL CORROSION:

1. Corrosion by Oxygen or Oxidation corrosion
2. Corrosion by other gases
3. Liquid Metal Corrosion

B. WET OR ELECTROCHEMICAL CORROSION

Electrochemical corrosion involves:

i) The formation of anodic and cathodic areas or parts in contact with each other

ii) Presence of a conducting medium

iii) Corrosion of anodic areas only and

iv) Formation of corrosion product somewhere between anodic and cathodic areas. This involves flow of electron-current between the anodic and cathodic areas.

TYPES OF ELECTROCHEMICAL CORROSION

The electrochemical corrosion is classified into the following two types:

(i) Galvanic (or Bimetallic) Corrosion

(ii) Differential aeration or concentration cell corrosion.

Galvanic Corrosion:

When two dissimilar metals (eg., zinc and copper) are electrically connected and exposed to an electrolyte, the metal higher in electrochemical series undergoes corrosion. In this

process, the more active metal (with more negative electrode potential) acts as anode while the less active metal (with less negative electrode potential) acts as cathode.
In the above example, zinc (higher in electrochemical series) forms the anode and is attacked and gets dissolved; whereas copper (lower in electrochemical series or more noble) acts as cathode.

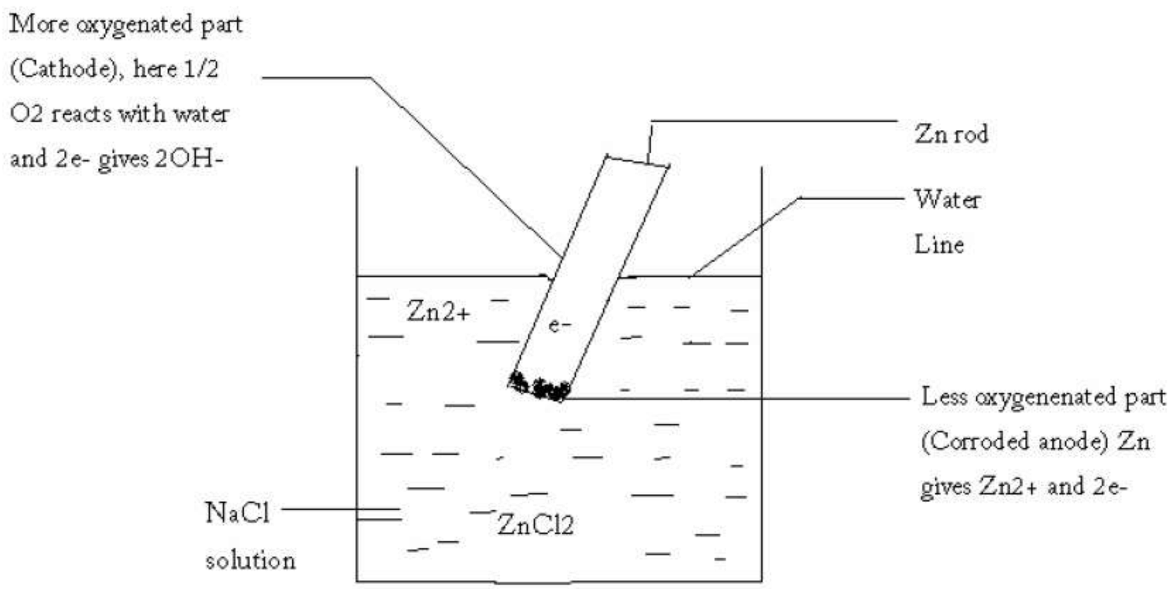
Mechanism:
In acidic solution, the corrosion occurs by the hydrogen evolution process; while in neutral or slightly alkaline solution, oxygen absorption occurs. The electron-current flows from the anode metal, zinc to the cathode metal, copper.
$Zn \rightarrow Zn^{2+} + 2e^-$ (Oxidation)
Thus it is evident that the corrosion occurs at the anode metal; while the cathodic part is protected from the attack.

Example: (i) Steel screws in a brass marine hardware (ii)Lead-antimony solder around copper wise; (iii) a steel propeller shaft in bronze bearing (iv) Steel pipe connected to copper plumbing.

Concentration Cell Corrosion:
It is due to electrochemical attack on the metal surface, exposed to an electrolyte of varying concentrations or of varying aeration. It occurs when one part of metal is exposed to a different air concentration from the other part. This causes a difference in potential between differently aerated areas. It has been found experimentally that poor-oxygenated parts are anodic.
Examples: i) The metal part immersed in water or in a conducting liquid is called water line corrosion.
ii) The metal part partially buried in soil.

Explanation: If a metal is partially immersed in a conducting solution the metal part above the solution is more aerated and becomes cathodic. The metal part inside the solution is less aerated and thus becomes anodic and suffers corrosion.

At anode: Corrosion occurs (less aerated) $M \rightarrow M^{2+} + 2e^-$
At cathode: OH- ions are produced (more aerated) $\frac{1}{2} O_2 + H_2O + 2e^- \rightarrow 2OH^-$



Examples for this type of corrosion are

1) Pitting or localized corrosion
2) Crevice corrosion
3) Pipeline corrosion
4) Corrosion on wire fence

Pitting Corrosion:

Pitting is a localized attack, which results in the formation of a hole around which the metal is relatively unattacked.

The mechanism of this corrosion involves setting up of differential aeration or concentration cell. Metal area covered by a drop of water, dust, sand, scale etc. is the aeration or concentration cell.

Pitting corrosion is explained by considering a drop of water or brine solution (aqueous solution of NaCl) on a metal surface, (especially iron). The area covered by the drop of salt solution as less oxygen and acts as anode. This area suffers corrosion, the uncovered area acts as cathode due to high oxygen content. It has been found that the rate of corrosion will be more when the area of cathode is larger and the area of the anode is smaller. Hence there is more material around the small anodic area results in the formation hole or pit.

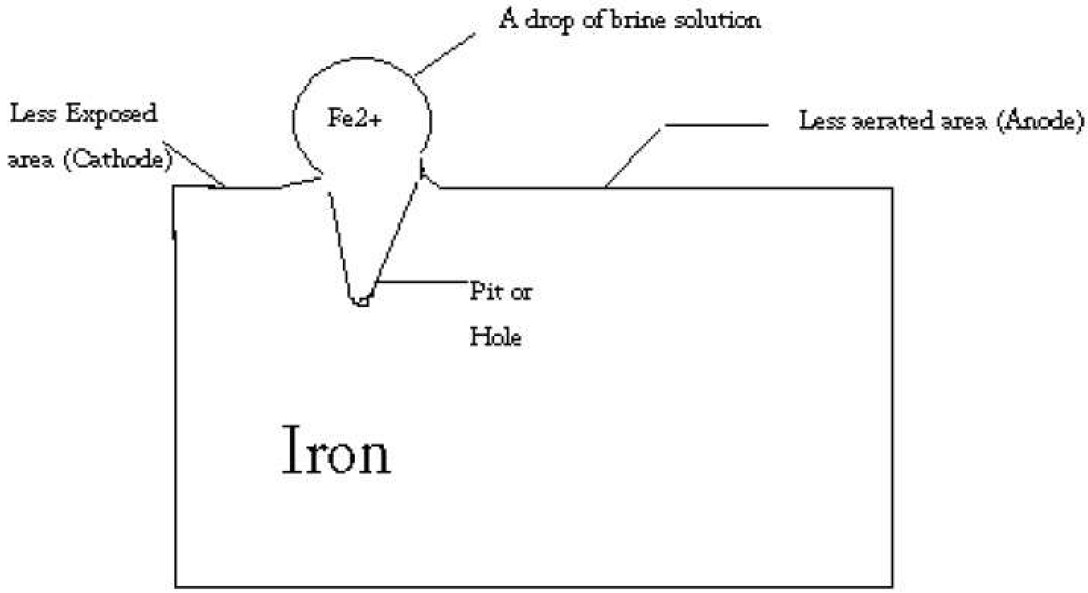
At anode: Fe is oxidized to Fe^{2+} and releases electrons.

$$Fe \rightarrow Fe^{2+} + 2e^-$$

At cathode: Oxygen is converted to hydroxide ion

$\frac{1}{2} O_2 + H_2O + 2e^- \rightarrow 2OH^-$

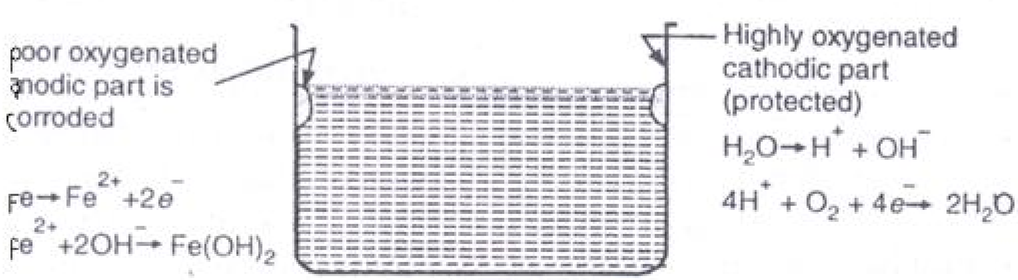
$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$



Waterline Corrosion:

Waterline corrosion is a type of oxidation process that can happen to materials in contact with water. Waterline corrosion occurs when one portion of a base material is submersed in the water and another portion is in contact with the air. This creates a differential of the amount of oxygen in contact with the material's surface above and below the waterline and results in a corrosive reaction.

Waterline corrosion occurs because of the differential in oxygen concentration between the atmosphere and the water. This happens because the portion of the substrate exposed to higher amounts of oxygen (the area exposed to air) becomes a cathode, while the portion of the substrate exposed to less oxygen (the area in contact with water) becomes an anode. The creation of an anode and a cathode allows oxidation to occur. This ultimately causes the area of the substrate submersed in water to become oxidized and corrode.



Waterline corrosion

FACTORS INFLUENCING CORROSION:

There are two factors that influence the rate of corrosion.

1. Nature of the metal 2. Nature of the corroding environment

(a) Nature of the metal: The rate of corrosion is influenced by physical state of the metal (such as grain size, orientation of crystals, stress, etc). The smaller the grain size of the metal or alloy, the greater will be its solubility and hence greater will be its corrosion.

b) **Purity of metal:** Impurities in a metal cause heterogeneity and form minute/tiny electrochemical cells (at the exposed parts), and the anodic parts get corroded.

c) **Over voltage:** The over voltage of a metal in a corrosive environment is inversely proportional to corrosion rate.

d) **Nature of surface film:** In aerated atmosphere, practically all metals get covered with a thin surface film (thickness=a few angstroms) of metal oxide. Greater the specific volume ratio, lesser is the oxidation corrosion rate.

e) **Relative areas of the anodic and cathodic parts:** When two dissimilar metals or alloys are in contact, the corrosion of the anodic part is directly proportional to the ratio of areas of the cathodic part and the anodic part.

f) Position in galvanic series:

g) Passive character of metal:

h) Solubility of corrosion products:

i) Volatility of corrosion products:

Nature of the Corroding Environment:

a) Temperature: The rate of corrosion is directly proportional to temperature ie., rise in temperature increases the rate of corrosion.

b) Humidity of air: The rate of corrosion will be more when the relative humidity of the environment is high.

c) Presence of impurities in atmosphere: Atmosphere in industrial areas contains corrosive gases like CO_2, H_2S, SO_2 and fumes of HCl, H_2SO_4 etc. In presence of these gases, the acidity of the liquid adjacent to the metal surfaces increases and its electrical conductivity also increases, thereby the rate of corrosion increases.

d) Presence of suspended particles in atmosphere: In case of atmospheric corrosion: (i) if the suspended particles are chemically active in nature (like NaCl, Ammonium sulphate), they absorb moisture and act as strong electrolytes, thereby causing enhanced corrosion.

e) Influence of pH: Generally acidic media (ie., pH<7) are more corrosive than alkaline and neutral media.

f) Nature of ions present:

g) Conductance of the corroding medium:

h) Formation of oxygen concentration cell:

(i)Flow velocity of process stream:

j) Polarization of electrodes:

CORROSION CONTROL (PROTECTION AGAINST CORROSION)

As the corrosion process is very harmful and losses incurred are tremendous, it becomes necessary to minimize or control corrosion of metals. Since the types of corrosion are so numerous and the conditions under which corrosion occurs are so different, diverse methods are used to control corrosion.

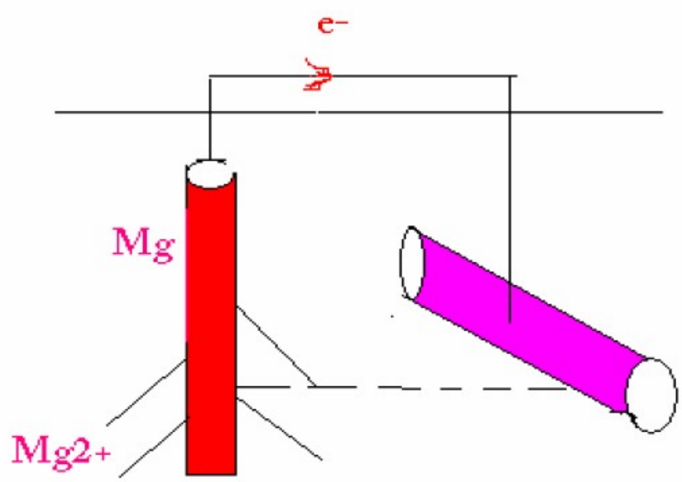
1. CATHODIC PROTECTION:

The reduction or prevention of corrosion by making metallic structure as cathode in the electrolytic cell is called cathodic protection. Since there will not be any anodic area on the metal, corrosion does not occur. There are two methods of applying cathodic protection to metallic structures.

a) Sacrificial anodic protection (galvanic protection)
b) Impressed current cathodic protection

a) Sacrificial anodic protection (galvanic protection)

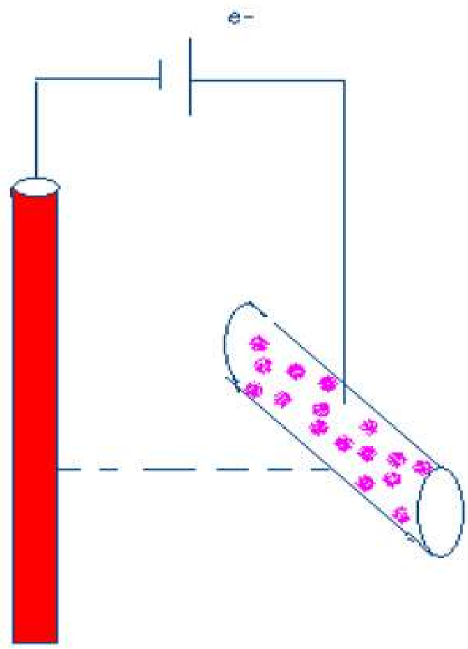
In this method, the metallic structure to be protected is made cathode by connection it with more active metal (anodic metal). Hence, all the corrosion will concentrate only on the active metal. The parent structure is thus protected. The more active metal so employed is called sacrificial anode. The corroded sacrificial anode block is replaced by a fresh one. Metals commonly employed as sacrificial anodes are magnesium, zinc, aluminium and their alloys.



In cathodic protection, an anode of a more strongly reducing metal is sacrificed to maintain the integrity of the protected object (eg., a pipeline, bridge, ship hull or boat).

b) Impressed current cathodic protection

In this method, an impressed current is applied in opposite direction to nullify the corrosion current and convert the corroding metal from anode to cathode. Usually the impressed current is derived from a direct current sources (like battery or rectifier on AC line) with an imsoluble, inert anode (like graphite, scrap iron, stainless steel, platinum or high silica iron). Impressed current cathodic protection has been applied to open water box coolers, water tanks, buried oil or water pipes, condensers, transmission line towers, marine piers, laid up ships etc. This kind of protection technique is particularly useful for large structures for long term operations.



In Impressed-current cathodic protection, electrons are supplied from an external cell so that the object itself becomes cathodic and is not oxidized.

2. PROTECTIVE COATINGS:

In order to protect metals from corrosion, it is necessary to cover the surface by means of protective coatings. These coatings act as a physical barrier between the coated metal surface and the environment.

Examples of organic coatings : Paints, Varnishes, Lacquers and Enamels

METALLIC COATINGS:

Corrosion of metals can be prevented or controlled by using methods like galvanization, tinning, metal cladding, electroplating, cementation, anodizing, phosphate coating, enamelling, electroless plating. Some of the methods are

Galvanizing: It is the process of coating iron or steel sheets with a thin coat of zinc to prevent them from rusting.

Tinning: It is a method of coating tin over the iron or steel articles.

ELECTROPLATING OR ELECTRODEPOSITION:

Electroplating is a coating technique. It is the most important and most frequently applied industrial method of producing metallic coating.

Electroplating is the process by which the coating metal is deposited on the base metal by

passing a direct current through an electrolytic solution containing the soluble salt of the coating metal.
The base metal to be plated is made cathode whereas the anode is either made of the coating metal itself or an inert material of good electrical conductivity (like graphite).

Passivity

Passivity refers to a corrosion preventative mechanism whereby an oxidation layer forms a continuous film on a metal's surface that prevents further corrosion.

Passivity of a metal is due to the formation of a highly protective, but very thin film on the surface of the metal under particular environmental conditions. Metals like chromium, aluminium, nickel form insoluble oxide films on their surface in presence of oxygen and hence are rendered passive.

Difference between (dry) chemical and (wet) electrochemical corrosion:

Sl. No.	Chemical Corrosion	Electrochemical Corrosion
1	It occurs in dry condition.	It occurs in the presence of moisture or electrolyte.
2	It is due to the direct chemical attack of the metal by the environment.	It is due to the formation of a large number of anodic and cathodic areas.
3	Even a homogeneous metal surface gets corroded.	Heterogeneous (bimetallic) surface alone gets corroded.
4	Corrosion products accumulate at the place of corrosion	Corrosion occurs at the anode while the products are formed elsewhere.
5	It is a self controlled process.	It is a continuous process.
6	It adopts adsorption mechanism.	It follows electrochemical reaction.